BTS-I(R/S)/II(S)-11.18-1908 Reg.No. [ | | | | | | | | ]

**A**

# B.Tech. Degree I Semester Regular and Supplementary/ II Semester Supplementary Examination November 2018

**CS/EC/IT GE 15-1104 B & CE/EE/ME/SE GE 15-1204 A**
**BASIC ELECTRICAL ENGINEERING**
(*2015 Scheme*)

Time : 3 Hours Maximum Marks : 60

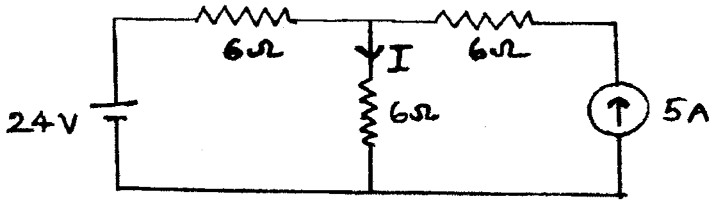
## PART A
(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) State Norton's theorem.

(b) Derive an expression for the energy stored in a capacitor of 'C' Farads when charged to a potential difference of 'V' volts.

(c) Calculate the current 'I' in the network using mesh analysis.




(d) Compare electric and magnetic circuits.

(e) Explain Faraday's law of electromagnetic induction and Lenz's Law.

(f) The voltage across and current through a circuit are given by

$v = 250\sin\left(314t - 10^\circ\right)$ ; $i = 10\sin\left(314t + 50^\circ\right)$

Calculate the impedance, resistance and reactance of the circuit.

(g) Prove that in a purely inductive circuit the current lags behind the voltage by an angle of 90° and the average power consumed is zero.

(h) Explain the advantages of three phase system compared to single phase system.

(i) A 220V DC series motor draws a current of 20A. The armature resistance is 0.1Ω and series winding resistance is 1.2Ω. Find the back emf.

(j) What are the different types of losses in a transformer? How can these losses be reduced?

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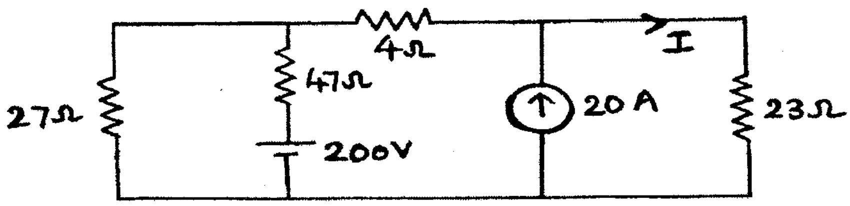
## PART B

(4 × 10 = 40)

II. (a) Calculate the current 'I' using Thevenin's equivalent circuit. (6)



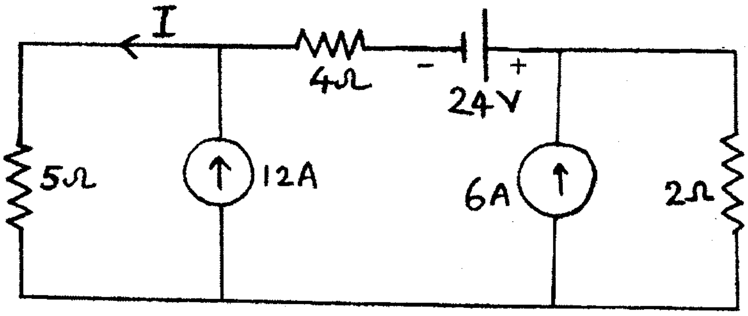

(b) Two capacitors, $A$ and $B$, having capacitances of $20\,\mu F$ and $30\,\mu F$ respectively, are connected in series to a $600\ V\ dc$ supply. Determine the potential difference across each capacitor. If a third capacitor $C$ is connected in parallel with $A$ and it is then found that the potential difference across $B$ is $400V$, calculate the capacitance of $C$ and the energy stored in it. (4)

**OR**

**(P.T.O.)**

III. (a) State and explain Superposition theorem. (4)

(b) For the circuit shown, use superposition theorem to calculate the value of current $I$. (6)




IV. (a) Explain the working of an energy meter with neat sketches. (6)

(b) Define mutual inductance in a magnetic circuit. Derive an expression for coefficient of coupling. (4)

**OR**

V. (a) Define (i) MMF (ii) Magnetizing force (iii) flux density (iv) reluctance (4)

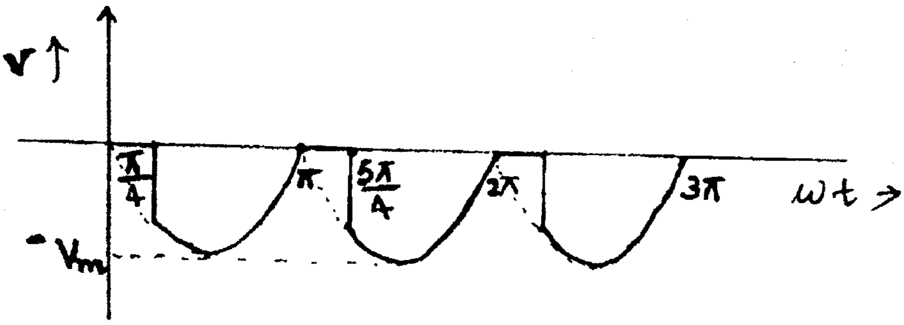
(b) A steel ring of 25 cm diameter and of circular cross section 3 cm in diameter has an air gap of 1.5 mm length. It is wound uniformly with 750 turns of wire carrying a current of 2.1 A. Calculate (i) m.m.f. (ii) flux density in air gap (iii) magnetic flux (iv) relative permeability of steel ring. Assume that iron path takes about 35% of total magneto motive force. (6)

VI. (a) A 10 Ω resistor and 400 $\mu$F capacitor are connected in series to a 240 V sinusoidal ac supply. The circuit current is 5 A. Calculate the supply frequency and phase angle between current and voltage. (4)

(b) Three inductive coils, each with a resistance of 22 Ω and an inductance of 0.05 H are connected (i) in star (ii) in delta, to a three phase 415 V, 50 Hz supply. Calculate for each of the above cases (i) phase current and line current (ii) total power absorbed. (6)

**OR**

VII. Find the form factor and peak factor of the given altering waveform. (10)




VIII. With a neat schematic, explain the working of a thermal power plant. (10)

**OR**

IX. (a) With a neat sketch explain the constructional details of a DC Machine. (6)

(b) Derive an equation for back emf in DC motor. (4)

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